Article

International Journal of Christianity & Education
2023, Vol. 27(3) 319–333

DOI: 10.1177/20569971221136551
journals.sagepub.com/home/ice
Sage

Faith integration in STEM courses for undergraduates: Exemplars of pedagogical practices

David Klanderman
Department of Mathematics and Statistics, Calvin University, Grand Rapids, MI, USA

Clayton Carlson
Department of Biology, Trinity Christian College, Palos Heights, IL, USA

Tina Decker
Department of Nursing, Trinity Christian College, Palos Heights, IL, USA

Fred Haan
Department of Engineering, Calvin University, Grand Rapids, MI, USA

Victor Norman
Department of Computer Science, Calvin University, Grand Rapids, MI, USA

Abbie Schrotenboer
Department of Biology, Trinity Christian College, Palos Heights, IL, USA

Derek Schuurman
Department of Computer Science, Calvin University, Grand Rapids, MI, USA

James Turner
Department of Mathematics and Statistics, Calvin University, Grand Rapids, MI, USA

Abstract
In his book *On Christian Teaching*, David Smith offers multiple ways in which one's Christian faith commitments influence pedagogical practices. Specific examples are drawn

Corresponding author:
David Klanderman, Department of Mathematics and Statistics, Calvin University, 3201 Burton Street SE, Grand Rapids, MI 49546, USA.
Email: dave.klanderman@calvin.edu

from the humanities, particularly from languages and literature. Some may conclude that such discussions are less appropriate in areas such as science, technology, engineering, and mathematics (STEM). We disagree and offer exemplars aligned with a set of six curricular criteria identified by Wayne Au. We argue that these exemplars emerge naturally from our Christian beliefs and are "faith-influenced" even if not every strategy embodies a narrower integration of faith and pedagogy.

Keywords
pedagogy, faith-influenced, science, technology, engineering, and mathematics disciplines, teaching and learning

Background and motivation for study

In his recent book *On Christian Teaching*, David Smith (2018) identifies three facets that can help teachers reframe and revise their teaching and learning strategies to better align with their Christian faith. The facets include seeing anew, choosing engagement, and reshaping practice. Seeing anew emerges from a creative imagination that fosters a greater focus on the context surrounding a lesson and its learning objectives. Choosing engagement calls for a greater sense of community and a more active role for students in the learning process. Reshaping practice impacts specific pedagogical choices that help to connect this new "sight" with the intentional choice to emphasize different approaches to engage the teacher and the students in the learning process. Smith then describes a number of specific examples from classroom teaching. In one such example, he describes a nine-minute portion of his initial class of the semester for his second-year German course. Through multiple phases, Smith delineates ways that he fosters a hospitable learning environment by having students first speak to introduce themselves to a partner, listen to their partner's introduction, and eventually introduce their partner to other students in their seating cluster. Consistent with other scholarship in this domain, the majority of the specific examples are drawn from the humanities, and in this case, from languages and literature in particular.

Other researchers have sought to extend the notion of faith-influenced pedagogical practices to these and related disciplines. For example, Gutacker et al. (2019) offer examples from history, English composition, political science, and theology. They build on the work edited by Smith and Smith (2011), in particular the writing of Paul Griffiths (2011) and his notion of liturgical practices within the classroom environment. Christian themes including hospitality, humility, confession, and catechesis are applied to specific pedagogical practices in these disciplines.

Some researchers have published work related to faith-influenced pedagogical practices in science, technology, engineering, and mathematics (STEM). Meehan (2007) offers the Shared Praxis method developed by Thomas Groome as a pedagogical strategy for teaching biology that is consistent with our Christian commitments. She calls us to "take equal care of who we are teaching, where this takes place, and how we go about it"

(p. 281). She suggests that Shared Praxis teaches wisdom for life no matter the content of the lesson. In computer science, Crisman (2015) offers four themes to connect pedagogy and the open source movement to Christian commitments, including stewardship of resources, building community, helping others, and creativity. Kihlstrom (2006) also emphasizes the concept of building community in her larger discussion of a computer science program. Norman (2015) discusses the importance of teaching virtues in a computer programming course, including hospitality, humility, integrity, honesty, creativity, stewardship, and diligence.

In engineering, Tipton (2019) suggests a reflective practice to allow not only for the integration of faith with content but also with learning. He pushes for connections between content and broader worldview and theological convictions. Brue (2017) advocates for the use of narrative as both genuinely Christian pedagogy and direly missing in engineering. He writes, "We want men and women to dream about doing engineering, but we give them nothing to dream about that transcends themselves or their profession. They are given derivations rather than story, proof rather than poetry, and empirical correlations rather than myths. It is no wonder we cannot retain our students as engineers" (p. 169). We argue that the same reasoning would apply to other STEM disciplines.

Finally, in mathematics, Eggleton (2019) calls for professors to infuse their teaching with spiritual and disciplinary discipleship as they seek to foster growth in their students' understanding of mathematics. He notes that "the spiritual discipleship of the student actually enhances the disciplinary discipleship and vice versa, allowing students to accept more life-impacting understandings of the content" (p. 44). Zonneveld (2015) notes that "Christian educators do not hold a monopoly on good teaching, as many unbelievers also display strong teaching qualities through common grace" (p. 124). She offers examples that recontextualize mathematics in ways that highlight Christian faith, such as an exploration of unrealistic body proportions in Barbie dolls and action figures as contrasted with the beauty given by God to each person. Wilkerson (2015) advocates a focus on helping students to value mathematics in his article discussing the cultivation of mathematical affections.

Analyzing survey responses from various schools within the Council of Christian Colleges & Universities (CCCU) suggests that professors specializing in computer science, math, and engineering were the least likely to integrate faith into their teaching, whereas professors specializing in religion and philosophy were most likely to do so (Kaul et al., 2017). Although there has been work in the area of faith integration as indicated above, these survey data may lead some to conclude that discussions of the integration of Christian faith with teaching and learning are less appropriate or even less possible in areas such as STEM. We disagree with this conclusion and propose to describe a series of exemplars from STEM disciplines including biology, computer science, mathematics, nursing, and engineering. Rather than publishing such faith-influenced pedagogical strategies in our separate disciplinary journals, we seek to provide an interdisciplinary analysis of pedagogical practices that demonstrate commonalities throughout the STEM disciplines.

Exemplars of faith-influenced pedagogical practices

To organize our presentation, we use the six pedagogical criteria identified by Wayne Au (2012) in his book *Critical Curriculum Studies: Education, Consciousness, and the Politics of Knowing*. These six criteria include physical space, use of language, behavioral practices, the role of time, aesthetics and creativity, and the role of power. Au builds on the prior work of Dwayne Huebner (1996) who had identified three of the categories. While Au's work does not emerge from a Christian perspective, the present paper seeks to describe pedagogical practices which could be classified as faith-influenced, or perhaps "faith-inflected" as defined by Robert Sweetman (2016). Each of us has taught at a Christian college or university for multiple years, some for multiple decades, and this research represents an attempt to identify and describe a set of pedagogical practices that align with our underlying Christian faith commitments. We found Au's categories to be a useful conceptual framing for our work, and we argue that we can apply this framework for the purposes of articulating these faith-influenced teaching strategies. We also note that while most of the authors find their home in reformed Christian intellectual and theological traditions, we argue that the strategies discussed below are not unique to a specific denomination and can be embraced by most if not all Christians. We view our research as building on the efforts discussed earlier but treating them as one cohesive discussion of the STEM disciplines. This section identifies multiple specific faith-influenced pedagogical practices for each of these six categories that have been implemented by faculty colleagues in the STEM disciplines. Some categories offer brief illustrations while others provide more detailed examples. As Christians teaching at faith-based colleges and universities, we seek to extend hospitality to all of our students and choose pedagogical strategies that create this environment. This theme permeates many of the exemplars discussed in the following sections.

The physical layout of the classroom environment

Au's first category, the physical environment of the classroom, "includes books, media, furniture, equipment, and even the architectural structure of the building" (p. 35). This physical environment category aligns with the "choosing engagement" facet cited by Smith (2018) in that some seating arrangements are more likely to foster this engagement. Physical space in mathematics classes has historically meant classrooms with desks in rows and columns all facing a blackboard. David Smith notes that the physical arrangement of desks and chairs has implications for the classroom community and the communication between students as well as between the teacher and students. In his mathematics education classes, Dave Klanderman changed the default arrangement of tables and chairs in a classroom from rows facing the front to clusters of four students each arranged for viewing access to both a screen and blackboard on adjoining walls. This initiative required the endorsement of several colleagues in the mathematics and statistics department who also taught classes in this room. A few decided to retain traditional lecture formats with the revised cluster seating while others adjusted teaching practices to leverage the student clusters with small group activities and discussions. Even in more lecture-centric mathematics classes, students were able to converse with neighbors more easily about a specific mathematical problem, thus fostering a greater sense of community. In his book describing many themes leading to human flourishing through mathematics, Francis Su writes “None of us can flourish without a supporting community – people with whom we can share joys and sorrows, hopes and fears. A community helps us normalize struggle, and realize ‘I’m not alone in my struggle’” (Su, 2020: 188). Cluster seating and related learning activities help embody this notion of human flourishing through hospitality, community, and collaboration.

Those teachers who embraced a more student-centered learning environment in this remodeled classroom space were able to leverage greater student-to-student, small group, and whole-class interactions. A colleague who teaches calculus in the same room noted that he was more willing to implement learning activities that shifted the emphasis from the instructor to the student in this physical space. These activities, such as a small-group analysis of a given graph of a function and its derivative, were addressed in a faculty development workshop for calculus professors hosted at Calvin University during Summer 2018. Other colleagues, including Derek Schuurman in his computer science capstone course, have been able to achieve the same pedagogical goal by relocating from a rows-and-columns classroom to one with more flexible seating patterns. In his team-taught mathematics capstone course at Trinity Christian College prior to coming to Calvin University, Klanderman and his colleagues held weekly book discussions with senior mathematics majors seated around a table in the bookstore café on campus. This informal setting in which all could see and interact with each other promoted a lively exchange of ideas related to a pair of books chosen to highlight Christian and other worldviews applied to mathematical ideas and practices (Klanderman and Robbert, 2012).

In his biology labs at Trinity Christian College, Clay Carlson utilizes seating at benches or tables for two or four students each, similar to many other laboratory-based science courses. He notes that these pairings naturally lead to lab partners, conversation partners, and later, homework partners. This collaboration also permeates the lecture sessions for the same course where he encourages students to sit near their partners, often at tables now formed into rows facing forward. This proximity helps these relationships formed during weekly laboratories to deepen. Likewise, Tina Decker led a redesign of nursing labs and classrooms centered on the idea of bringing students together and being more responsive to the needs of all students, including commuters. A nursing student lounge was added. The skills lab was remodeled to create workstations for students to work as partners. The new classroom is now set up with flexible seating arranged in a rectangle so that all participants can see each other. In all of the instances described above, the design or redesign of the physical environment to promote a learning community resonates with underlying Christian principles of hospitality and communal engagement.

The use of language

Au’s second category focuses on the language of discourse employed by both teachers and students within the classroom and applied to the specific academic discipline. This category connects to the “seeing anew” facet cited by Smith (2018) in that patterns of discourse can enhance the overall teaching and learning environment. In his computer programming courses, Vic Norman emphasizes the use of comments interspersed within lines of code as well as consistent names for variables and other program components. While other professors would endorse this practice, Norman's motivation is one of Christian hospitality. In particular, future users of a computer program will find it much easier to make modifications and improvements if the original programmer chooses to share some of her thinking that led to the original design (Norman, 2012). In a sense, programmers of today are anticipating a larger community in the future that will have the task of modifying and extending these computer programs. In the preface to the book, *Structure and Interpretation of Computer Programs*, the authors make the startling claim that "programs must be written for people to read, and only incidentally for machines to execute" (Ableson et al., 1996). This pedagogical approach emphasizes the role of hospitable language within the field of computer science in that the professor reminds students to consider future audiences when writing computer code.

In her biology course taught at Trinity Christian College, Abbie Schrotenboer argues that the use of language to discuss the environment can affect whether students view it simply as a resource to be consumed or as a world full of complex ecosystems and creatures designed by God. Inspired by the writing of Norman Wirzba (2015), she seeks to use the terminology of the earth as *creation* to highlight God's active role in the world, whereas speaking of *nature* or *the environment* can lead to a more disconnected view. For example, "natural resources" enforce a human-centric utility, whereas describing water and land as "provisions" reminds students that these are gifts offered by the Giver. Terms such as biodiversity may be less helpful when speaking to a general audience (such as in an introductory environmental science course), whereas using phrases like "plants and animals" or other specifics tend to draw students' attention to the particulars in the world around them.

Similarly, Carlson initially seeks to explain beauty in his microbiology courses without a heavy emphasis on terminology. Later, once students see the beauty, he can implement the use of more scientific language. Next, he invites students to practice this terminology both in writing prompts and verbal discussions with a student partner. Eventually, formal assessments are used to reinforce the use of technical language prior to starting the next instructional unit by building upon these formal terms to understand the next level of beauty. For example, students first learn some bacterial cells are round before describing these cells as cocci in order to work toward understanding the meaning of streptococci. Carlson describes this entire pedagogical practice as one of invitation, formation, and development in order to discover more and more beauty within the discipline of microbiology.

Behaviors of students, teachers, and others in the educational environment

Au's third category highlights "the behaviors of the people in the [learning] environment (students, teachers, and other school personnel)" (p. 35). This category aligns with the "choosing engagement" facet cited by Smith (2018) in that the behaviors by all participants impact the effectiveness of the learning environment. In his engineering courses at Calvin University, Fred Haan emphasizes the practice of lifting up one's eyes to see the connections that a given activity has to other people. In engineering—and engineering education in particular—it is often too easy to develop solutions that are technically correct but do not fully appreciate human impacts. How will a design affect the person who will fabricate it? How will it affect the person who will use it? Haan highlights the three-way communication among engineering design team, fabrication team, and client as a critical step in the overall design process. The regular consideration of such questions is a behavioral practice in Au's list of pedagogical criteria. Such an approach finds resonance with the notion of a Christian community in which members not only consider the interests of one another but also provides a concrete opportunity to love our neighbor.

In her nursing clinical rotations at Trinity Christian College, Decker used the recent pandemic as an opportunity to refocus her students' attention to physical, mental, and spiritual wellness. Responding to the overwhelming mental health needs within the nursing profession, including burnout and compassion fatigue that have been documented at unprecedented rates, Decker decided to reassign some clinical hours to student wellness. She argues that teaching students how to meet the mental health needs of others while potentially ignoring their own is problematic. This innovation emerged from insights of alumni serving on a panel who related their personal struggles in these areas. Decker then trained her students to assess their own wellness needs and to identify and implement possible interventions. Students are then prompted to reflect on the impact of the interventions and continue to seek out activities that are responsive to their wellness needs.

Schrotenboer emphasizes the practice of hospitality in her environmental biology laboratories. She defines this practice as ensuring that everyone in a laboratory team is following along during the lab and has opportunities to contribute. Although this approach can be beneficial in any group work setting, Schrotenboer notes that this strategy is particularly relevant when students are juggling hands-on laboratory skills, engaging content, and working with their lab partners. Likewise, Carlson stresses the theme "all students do" in his microbiology labs, meaning that "no one watches" and "no one helps." Each student in the room is responsible to think through the experiments, carry out the work, and analyze the results. In addition, each student is responsible to ensure that those students nearby are able to think through, do, and analyze the experiments. Constant communication among lab team members is a necessary prerequisite to achieve these results, but this approach to learning also embodies a notion of Christian hospitality and concern for all members of the lab team. Carlson notes that students and professor alike are expected to ask questions, answer questions, and listen to input from others. These expectations are modeled during lecture sessions and embodied during subsequent laboratory sessions. The pedagogical approaches discussed in this section highlight the behaviors of students, teachers, and clients in classroom, clinical, and workplace settings.

The role of time

Au's fourth category relates to time, both in the sense of how it is used within the classroom setting and how the past, present, and future of the discipline can inform pedagogy. This issue connects to the “reshaping practice” facet cited by Smith (2018) in that teachers benefit from a careful analysis of the use of time in the teaching and learning setting. In traditional mathematics classes, some students struggle to complete tests within a specified time. This sometimes results in an increased sense of anxiety among students in mathematics courses as they constantly worry about having sufficient time to demonstrate their mastery of the material. In his content courses for elementary education majors, Klanderman decided to extend to all of his students the hospitality of an untimed test, previously accessible only to students with documented learning accommodations. The approach was made possible by reserving an additional room and allowing students to start early, finish late, or both. In rare cases, an alternate test window was necessary. By offering multiple pathways to extended time and by proactively reserving an auxiliary testing space, problems such as student schedule conflicts and available space were avoided. The initiative applies Au’s role of time to make the formal assessment experience a more hospitable one for his students. This pedagogical practice is influenced by a Christian sensibility that each of us has different God-given gifts and abilities, and temperaments, and we are called to serve and honor God in our callings. Klanderman’s decision to create temporal space to accommodate all learners and alleviate math test anxiety seeks to promote the optimal use of students’ gifts and talents in a mathematics classroom setting. In turn, these students reflect on these experiences in a subsequent math methods course and hopefully place a decreased emphasis on speed and an increased emphasis on conceptual learning and problem-solving in their future classrooms. In his biology courses, Carlson adopts a similar approach, giving students essentially unlimited time on exams. He notes that he has never taken a test from a student before she or he considered the test to be completely finished.

Schuurman requires his students to reflect on the time spent with digital devices by assigning a 24-hour “digital fast” to his capstone students. Students choose one day where they “give up” some tech device, app, or activity (e.g., social media, gaming, and texting). Students then submit a personal reflection on their experience of this “technology fast.” Students are prompted with questions such as how this activity made them feel, where their thoughts were turning to, and if they would consider making a technology fast part of their regular life rhythms. Through this practice, Schuurman hopes to nudge students to think not only about how compelling certain technologies can be but also about the place of sabbath and time in the lives of his students. This practice can also help future computer scientists to consider the impact that the design of software products has on users.

In her nursing courses, Decker built on the attention to physical, spiritual, and mental wellness described earlier to another one of her courses. She requires students to submit evidence of completing a physical wellness activity such as 10,000 steps or 30 min of moderate exercise on at least 5 days each week. In addition, students must document a mental or spiritual wellness activity on at least 5 days each week, although the specific activity can be selected by the student and is broadly defined. Decker notes that, as a Christian educator, this pedagogical approach conveys the message that she cares enough about her students as persons created in God’s image to require students to dedicate time each week to promote their physical, mental, and spiritual wellness. Students also came to recognize that they are created by God with intricate and amazing design, and they need to listen and be responsive to the feedback from their bodies. In these ways, time plays a key role in both formal assessment and ongoing self-care in these STEM classrooms.

The role of creativity and artistic elements in teaching

Au's fifth category identifies a creative or artistic element in teaching and ways that such focus impacts student learning. This category aligns well with the "seeing anew" facet identified by Smith (2018) in relation to the creative insights applied to teaching. In his 3-week intensive course at Calvin University designed to meet a core requirement called "Developing the Christian Mind," Jim Turner has organized the course around the theme of beauty in the context of a larger discussion of the development of a Christian world and life view, with an emphasis on reformed intellectual and theological traditions. Specifically, his goals include sharing what it means to think beautifully about mathematics through the construction of an elegant proof or an insightful problem-solving strategy, and applying aesthetics to the analysis of mathematical solutions to real-world problems. Turner has written and spoken at a greater length on the links between this pedagogical practice and the notion of beauty as found in the writings of St. Bonaventure and Thomas Aquinas (Turner, 2017, 2019), thus giving evidence that this link to a Christian perspective applies more broadly than a narrow focus on a reformed Christian interpretation. The aim was to provide a perspective of how humanity as *imago dei* functions as creative participants in the world created by the Triune God.

He also describes the work of David Mumford (2015) who articulates how beauty can be seen in mathematics through four different categories of mathematicians. Mumford defines explorers as those who generate new ideas or perspectives within the discipline, either by mapping out new mathematical territory as gem collectors who identify new mathematical objects and their features or by giving new perspectives to known objects via types of relations. Alchemists find connections between two previously disjoint areas of mathematics, such as analytic geometry which bridges the divide between algebra and geometry. Wrestlers focus on the relative sizes and strengths of mathematical objects to produce results such as the prime number theorem. Finally, detectives pursue clues and follow trails to answer deep and difficult questions, either uncovering a hidden layer in the solution of a larger problem or naming new objects to make explicit a key object (e.g., π) that was previously defined implicitly (i.e., the ratio of the circumference to the diameter of a circle). Taken together, these different types of mathematicians demonstrate how to think beautifully about mathematics. Each of these approaches are modes of understanding that fit under such overarching themes as unveiling deeply hidden patterns and discovering unexpected connections. Exploring how such modes of understanding operate within the context of mathematics provides a platform for further understanding of how they operate within the search for knowledge in general. As Christians teaching mathematics and other STEM disciplines, we seek to create learning opportunities for students to discover and describe these patterns and connections that occur within the discipline as well as within God's creation. Turner encourages all of his students, whether liberal arts majors taking a 3-week intensive course or advanced mathematics majors, to approach mathematics with a similar aesthetic.

Norman has explored the role of aesthetics in the creation of a computer program. In his prior work, he distinguishes features of “ugly” programs from those which expert computer programmers consider to be “beautiful” (Norman, 2012). He emphasizes the artistic component to the design of a computer program, including the analysis of a series of criteria proposed by Donald Knuth that help an independent observer to judge the beauty of a created computer program (Knuth, 1974). These criteria include utility, robustness, and efficiency, among others, and point to an aesthetic of creating beautiful computer programs.

In his microbiology courses, Carlson notes that the interrelated complexity of biology captures the attention of his students, whether or not he highlights this feature. From biochemical machines deep inside a cell reading DNA and writing RNA, to the connections between a brain and a muscle cell, to the dependence of the trees on the tiny soil microbes, biology is beautiful. Were these biological components merely machines, his students would be impressed. However, these biological wonders are alive! The tiniest cell is like a vast Rube Goldberg machine, and every time a student sees and experiences it, they respond with amazement. By naming and emphasizing the inherent beauty of these systems and structures, students experience and respond to the creative, the beautiful, and the artistic dimensions of the curriculum described by Au. They also emerge from Christian beliefs about the order, regularity, and beauty within God’s created world and the academic disciplines. Likewise, in her nursing health assessment course, Decker highlights the functioning of the human heart, how to think of all the precise actions that go into every beat, and how that ultimately leads to life that was so perfectly designed by the Creator. Students get to listen to the heart beat in a whole new way when called to think about what it actually represents.

The assignment of power roles within a learning community

Au’s sixth and final category highlights power roles within the teaching and learning environment as well as issues related to the resolution of conflicts and disagreements that may arise within the classroom. This category connects to the “choosing engagement” facet cited by Smith (2018) in that it focuses on the engagement rules by teachers and students in the classroom. In the area of power in the classroom, the traditional model locates the power with the teacher while students primarily respond to question prompts. In Klanderman’s mathematics education classes, he assigns regular readings. However, unlike the standard pedagogical technique of the teacher leading the discussion, he periodically assigns primary discussion roles to either individual students or small clusters of students. In this way, power is transferred to the students during this portion of the class sessions, applying yet another criterion described by Au.

Klanderman frequently assigns a small group of students to serve as the leaders of class discussion for readings (Klanderman, 2001). The shift of authority in class discussions is even more widespread in courses that focus on methods of teaching mathematics at the elementary, middle school, and high school levels. Students unpack assigned readings through syntheses written prior to class and discussion of key ideas and specific sample learning activities during the class discussion led by these small groups. Where appropriate, these students also reflect on different pedagogical strategies through the lens of their own prior school experiences and concurrent field placements. He cedes the power of the learning environment from one of the "sage on the stage" to more of a "guide on the side." In some areas such as differentiated instructions and strategies for teaching students with special needs, the students enrolled in the class who are completing special education majors are well-placed to offer unique insights to the classroom discussion of these topics. For example, several best practices for assisting students with moderate to severe learning challenges in a mathematics classroom overlap with similar techniques for other disciplines that these special education majors have already learned. As one example, providing a graphic organizer to assist a student with reading challenges can apply to mathematical as well as scientific readings. This pedagogical practice can trace its roots to the Apostle Paul who exhorts Christians to value each member's own unique gifts as part of a larger body (I Corinthians 12: 1–31).

In her biology courses, Schrotenboer allows students a degree of choice in their course projects and lab experiments. From the development of the initial question through the analysis of the collected data, students feel empowered to direct their curiosity into their learning. This shift of power increases their excitement, engagement, and ultimately their learning. Schrotenboer teaches a conservation biology course in which students complete a project designed to help them learn and share about God's creation on campus and in the local area by choosing the specific focus. Past projects have ranged from birds to fungi to a tour on a campus trail along a local creek. Results of these projects have included reference guides, pamphlets, and a computer program to test one's knowledge. These pedagogical approaches that empower students also enable them to learn how science is actually done.

All of the abovementioned exemplars demonstrate how STEM disciplines provide conducive environments and contexts in which Christian commitments can influence pedagogical practices. While other professors, even those with different or no faith commitments, may use one or more of these same pedagogical practices, the authors point out that these practices are consonant with and emerge naturally from our Christian faith commitments. This observation is consistent with the view of Nicholas Wolterstorff (1984) who writes "In some cases a Christian scholar and a non-Christian scholar may each justifiably accept a particular scientific theory. For a given theory may align with the control beliefs of the non-Christian scholar and also with the belief-content of the authentic commitment of the Christian scholar" (p. 83). We contend that the motivations for pedagogical approaches are similarly affected by the belief-content of our authentic Christian commitments and our desire to create learning spaces characterized by hospitality and community.

Avenues for future scholarship

While the preceding discussion has offered multiple exemplars of faith-influenced pedagogical practices, there is much more that can be done in this area. First, others may be able to extend Au's framework by providing additional exemplars. We admit that Au is not the only possible conceptual framework for classifying faith-influenced pedagogical practices, even if we also see connections with the three facets described by Smith (2018). However, we found Au's six categories a useful way to connect the various teaching strategies used across these related disciplines. These potential extensions may come from the STEM disciplines or from other portions of the academic spectrum. In a sense, the present article presents an existence proof for such pedagogical practices. However, it makes no claim of uniqueness or sufficiency. Collecting data to assess the effectiveness of these pedagogical approaches, whether from student submitted work, recordings of class sessions, or course evaluations, would provide evidence to support the exemplars cited in this article as well as those proposed in the future. This future research would help move these faith-influenced pedagogical strategies from anecdotal to generalizable. Further, both learning outcomes and program outcomes for these STEM disciplines may need to be revised in light of these pedagogical strategies, placing a greater emphasis on the role of Christian values and their connections to both the teaching practices and the resulting student learning outcomes.

Other avenues for future scholarship in this area may consider frameworks other than the one articulated by Au. For example, Griffiths (2011) argues for the inclusion of liturgical practices in classroom settings. Some of these practices, such as hospitality and confession through humility, would apply in a broad spectrum of academic disciplines. Others, such as Cody Stecker's use of catechesis in a theology classroom (Gutacker et al., 2019), might have a narrower range of applicability. Still others have offered exemplars from a variety of disciplines, including the natural sciences (cf. D. Smith and J. Smith, eds, 2011). For that matter, one should be encouraged to explore more pedagogical practices that emerge out of David Smith's categories: seeing anew, choosing engagement, and reshaping practice.

Finally, the focus of the present study is at the undergraduate level. However, there is a similar need for faith-influenced pedagogical practices at the K-12 level. Some of the ideas discussed above, such as the use of physical space or the timeframe of testing, would apply equally well to these educational levels. Other ideas, such as hospitable and aesthetically pleasing computer code, might be of more limited use at the K-12 level. Perhaps a collaboration of teaching professionals from kindergarten through college would prime the pump to produce additional pedagogical practices that align with Christian beliefs and would apply to a variety of grade levels and disciplinary contexts.

Summary

This study identifies a collection of pedagogical practices employed in one or more of the STEM disciplines. We argue that these practices are consonant with our Christian faith commitments and have defined them as faith-influenced. Just as Robert Sweetman defines Christian scholarship as that which "is not so much about the scriptures as aligned to them" (p. 22), so we argue that our specific pedagogical choices align with our underlying faith commitments. George Marsden (1998) uses the phrase "intrinsically and yet not exclusively Christian" to describe his notion of Christian scholarship. We claim a similar lack of exclusivity to our pedagogical practices while noting that they emerge naturally from our faith commitments. We hope that colleagues will choose to join this conversation with additional exemplars from an even wider array of academic disciplines that potentially apply at all grade levels.

Acknowledgments

The authors gratefully acknowledge the support of the Calvin Center for Christian Scholarship for this research, including Dave Klanderman's participation in the Writer's Co-Op during June 2019 and a small grant awarded to Jim Turner and Dave Klanderman to work on this and related research projects. An earlier version of this research was presented at the Kuyers Conference held at Calvin University in October 2019.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the de Vries Institute during 2021 that provided Dave Klanderman the time necessary to complete this article.

ORCID iDs

David Klanderman https://orcid.org/0000-0002-1428-6084
Derek Schuurman https://orcid.org/0000-0001-5095-3799

References

Ableson H, Sussman GJ and Sussman J (1996) *Structure and Interpretation of Computer Programs*. Cambridge, MA: MIT Press.

Au W (2012) *Critical Curriculum Studies: Education, Consciousness, and the Politics of Knowing*. New York: Routledge.

Brue E (2017) The craft of storytelling in engineering education. In: *Proceedings of the Christian Engineering Conference*, Cedarville, OH, June 28-30, pp. 167–180.

Crisman K-D (2015) Open source software and Christian thought. *Perspectives on Science and Christian Faith* 67(1): 3–13.

Eggleton P (2019) Developing mathematicians: The benefits of weaving spiritual and disciplinary discipleship. In: *Proceedings of the Association of Christians in the Mathematical Sciences 22nd Biennial Conference*, Marion, Indiana, 2019, pp. 40–46.

Griffiths P (2011) Curiosity to studiousness: Catechizing the appetite for learning. In: Smith DI and Smith JKA (eds) *Teaching and Christian Practices: Reshaping Faith and Learning*. Grand Rapids, MI: William B. Eerdmans Publishing Company, pp. 102–122.

Gutacker P, Travers Parker E, Strecker C, et al. (2019) A symposium on teaching virtue: Interdisciplinary perspectives on pedagogy, liturgy, and moral formation. *International Journal of Christianity & Education* 23(2): 204–230.

Huebner D (1996) Teaching as moral activity. *Journal of Curriculum and Supervision* 11(3): 267–275.

Kaul CR, Hardin KA and Beaujean A (2017) Predicting faculty integration of faith and learning. *Christian Higher Education* 16(3): 172–187.

Kihlstrom K (2006) Asserting CS != can't socialize: Building community in a computer science program. *Journal of the Association of Christians in the Mathematical Sciences*. Available at: https://acmsonline.org/home2/wp-content/uploads/2016/05/Kihlstrom.pdf (accessed 7 June 2022).

Klanderman D (2001) Teaching and learning of mathematics: The influence of constructivism and a Christian response. In: Bradley J and Howell R (eds) *Mathematics in a Postmodern Age: A Christian Perspective*. Downers Grove, IL: InterVarsity Press, pp. 338–359.

Klanderman D and Robbert S (2012) Review essay of *Mathematics Through the Eyes of Faith* by James Bradley and Russell Howell and *A Certain Ambiguity: A Mathematical Novel* by Gaurav Suri and Hartosh Singh Bal. *Christian Scholar's Review* 41(4): 401–405.

Knuth D (1974) Computer programming as an art. *Communications of the ACM*, pp. 17(12): 667–673.

Marsden G (1998) What difference might Christian perspective make? In: Wells R (ed) *History and the Christian Historian*. Grand Rapids: William B. Eerdmans Publishing Company, pp. 11–22.

Meehan A (2007) Paradise regained: Teaching science from a Christian standpoint in a postmodern age. *Perspectives on Science and the Christian Faith* 59(4): 275–282.

Mumford D (2015) Math & beauty & brain areas. In: David Mumford: Archive for Reprints, Notes, talks, and blog. Available at: http://www.dam.brown.edu/people/mumford/blog/2015/MathBeautyBrain.html (accessed 9 September 2021).

Norman V (2012) Beauty and computer programming. *Inroads* 3(1): 46–48.

Norman V (2015) How will you practice virtue without skill? Preparing students to be virtuous computer programmers. *Journal of the Association of Christians in the Mathematical Sciences*. Available at: https://acmsonline.org/journal/journal-archives/2014-2015-journal/preparing-students-to-be-virtuous-computer-programmers/ (accessed 7 June 2022).

Smith D (2018) *On Christian Teaching: Practicing Faith in the Classroom*. Grand Rapids, MI: William B. Eerdmans Publishing Company.

Smith DI and Smith JKA (eds) (2011) *Teaching and Christian Practices: Reshaping Faith and Learning*. Grand Rapids, MI: William B. Eerdmans Publishing Company.

Su F (with reflections by Christopher Jackson) (2020) *Mathematics for Human Flourishing*. New Haven, CT: Yale University Press.

Sweetman R (2016) *Tracing the Lines: Spiritual Exercise and the Gesture of Christian Scholarship*. Eugene, OR: Wipf and Stock Publishers.

Tipton J (2019) Opportunities for faith & learning integration in a mechanical engineering instrumentation & measurement course. In: *Proceedings of the Christian Engineering Conference*, Sioux City, IA, July 11-13, 2019, pp. 22–33.

Turner J (2017) Seeing beauty in mathematics: On Bonaventure's 'reduction' of mathematics to theology. *Cithara: Essays in the Judeo-Christian Tradition* 57(1): 38–57.

Turner J (2019) Thinking beautifully about mathematics. In: *Proceedings of the Association of Christians in the Mathematical Sciences 22nd Biennial Conference*. Marion, Indiana, 2019, pp. 171–187.

Wilkerson J (2015) Cultivating mathematical affections: The influence of Christian faith on mathematics pedagogy. *Perspectives on Science and Christian Faith* 67(2): 111–123.

Wirzba N (2015) *From Nature to Creation: A Christian Vision for Understanding and Loving Our World*. Ada, MI: Baker Academic.

Wolterstorff N (1984) *Reason within the Bounds of Religion*. Grand Rapids, MI: William B. Eerdmans Publishing Company.

Zonneveld V (2015) Practical applications of an integrally Christian approach to teaching mathematics. *Perspectives on Science and Christian Faith* 67(2): 124–134.